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Course: Calculus Beta

Assignment: MyLab Math - Uge 9

Find the Taylor polynomials of orders 0, 1, 2, and 3 generated by f at a .

$$f(x) = e^{10x}, a = 0$$

Let f be a function with derivatives of order for $k = 1, 2, \dots, N$ in some interval containing a as an interior point. Then for any integer from 0 through N , the Taylor polynomial of order n generated by f at $x = a$ is the polynomial shown below.

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

The formula that corresponds to the Taylor polynomial of order 0 generated by f at $a = 0$ is shown below.

$$P_0(x) = f(0)$$

Find $f(0)$.

$$\begin{aligned} f(0) &= e^{10(0)} \\ &= 1 \end{aligned}$$

Therefore, the Taylor polynomial of order 0 generated by $f(x) = e^{10x}$ at $a = 0$ is shown below.

$$P_0(x) = 1$$

The formula that corresponds to the Taylor polynomial of order 1 generated by f at $a = 0$ is shown below.

$$P_1(x) = f(0) + f'(0)x$$

Find $f'(x)$.

$$f'(x) = 10e^{10x}$$

Now find $f'(0)$.

$$f'(0) = 10$$

Recall that $f(0) = 1$. Use $f(0)$ and $f'(0)$ to find the Taylor polynomial of order 1 generated by $f(x) = e^{10x}$ at $a = 0$.

$$\begin{aligned} P_1(x) &= f(0) + f'(0)x \\ &= 1 + 10x \end{aligned}$$

The formula for the 2nd order Taylor polynomial of f at $a = 0$ is shown below.

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$$

Recall that $f'(x) = 10e^{10x}$. What is $f''(x)$?

$$f''(x) = 100e^{10x}$$

Now find $f''(0)$.

$$f''(0) = 100$$

Recall that $P_1(x) = 1 + 10x$. Use $P_1(x)$ to find the Taylor polynomial of order 2 generated by $f(x) = e^{10x}$ at $a = 0$.

$$\begin{aligned} P_2(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 \\ &= 1 + 10x + 50x^2 \end{aligned}$$

The formula for the 3rd order Taylor polynomial of f at $a = 0$ is shown below.

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3$$

Recall that $f''(x) = 100e^{10x}$. What is $f^{(3)}(x)$?

$$f^{(3)}(x) = 1000e^{10x}$$

Now find $f^{(3)}(0)$.

$$f^{(3)}(0) = 1000$$

Recall that $P_2(x) = 1 + 10x + 50x^2$. Use $P_2(x)$ to find the Taylor polynomial of order 3 generated by $f(x) = e^{10x}$ at $a = 0$.

$$\begin{aligned} P_3(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 \\ &= 1 + 10x + 50x^2 + \frac{500}{3}x^3 \end{aligned}$$

The Taylor polynomials of orders 0, 1, 2, and 3 generated by $f(x) = e^{10x}$ at $a = 0$ are listed below.

$$P_0(x) = 1$$

$$P_1(x) = 1 + 10x$$

$$P_2(x) = 1 + 10x + 50x^2$$

$$P_3(x) = 1 + 10x + 50x^2 + \frac{500}{3}x^3$$